Sanduni Wathsala

SOFTWARE ENGINEER INTERN

Phone: (+94) 704587348 | **Email :**wathsalakjs.23@uom.lk

LinkedIn: linkedin.com/in/sanduniwathsala | **GitHub:** github.com/WathsalaKJ | **Address:** Colombo, Sri Lanka

ABOUT MYSELF

Motivated IT and Management undergraduate with a strong foundation in Java, object-oriented programming, data structures, and algorithms, and hands-on experience in full-stack web development using JavaScript, Node.js, React, Next.js, Python, Flask, and relational databases (MySQL, PostgreSQL). Experienced in RESTful API development, unit and integration testing, Agile/Scrum team environments, and applying machine learning concepts to real-world data problems, including embedded systems and IoT hardware-software integration. Skilled in building scalable applications with a strong focus on user experience and data management. Proven ability to collaborate in cross-functional teams and contribute to both frontend interfaces and backend logic. Passionate about creating impactful solutions and continuously learning new tools and technologies.

EDUCATION

University of Moratuwa *2024 - 2028*

BSc (Hons) Information Technology and Management, Faculty of Information Technology

SKILLS

- **Languages:** Python, Java, C, SQL, JavaScript, TypeScript, PHP
- **Frameworks & Libraries:** React, React Native, Next.js, Node.js, Flask, FastAPI, Tailwind CSS, scikit-learn
- **Databases & Cloud:** MySQL, PostgreSQL, Supabase, Firebase, Firestore
- **Tools & Platforms:** Git, GitHub, Docker, Vercel, Postman, REST APIs, JWT Authentication, AWS

PROJECTS

Digital Employee Performance Management System (EPMS) | GitHub Link *2025-2026*

Group Project | React, Next.js, TypeScript, Python Flask, Tailwind CSS, Supabase (PostgreSQL)

- Built a role-based Performance Management System for Dart Global Logistics Inc. (Colombo), an aviation and logistics company operating across 25 countries and numerous branches, department, and sub-department levels, enabling KPI evaluation and potential assessment across a six-tier organizational hierarchy
- Key features developed: Reports Module (frontend and backend), Potential Assessment workflow (self-assessment, supervisor review, system-calculated Overall Pillar Rating), and role-based dashboard UI components, structured using object-oriented design principles for maintainability across modules
- Built a live bell curve analytics engine, replacing static data with real-time performance calculations scoped by organizational role, while working in an Agile/Scrum team environment with sprint planning and stand-ups
- Wrote unit and integration tests (Jest, pytest) to validate module logic and API endpoints before merge

EcoSync - Home Solar Panel Output Forecaster | GitHub Link *2026- Ongoing*

Group Project | React, Next.js, Python FastAPI, PostgreSQL, scikit-learn

- Built a live weather-data pipeline (FastAPI, REST API, PostgreSQL) with scheduled refresh and trained Random Forest model on real solar generation data, reducing prediction error from 38.8% to 14.4% MAPE
- Engineered a capacity-normalized training approach so predictions generalize across solar system sizes despite training on utility-scale plant data

- Implemented JWT-based authentication and RESTful backend endpoints, and wired the Next.js frontend to consume the forecast API
- Collaborated across a Forecast/ML vertical and a User/Personalization vertical, integrating a correction-factor system that adjusts predictions based on each user's logged actual output

Snapense - AI-Powered Expense Tracker | GitHub Link *2026*
Individual Project | React Native, Flask, PostgreSQL, Google Gemini API

- Built a full-stack mobile expense tracker with Flask/PostgreSQL API and React Native frontend, with JWT auth and a normalized relational schema
- Wrote pytest unit tests for API endpoints and anomaly-detection logic to catch regressions
- Integrated a provider-agnostic vision LLM pipeline to auto-extract and categorize receipt data from photos
- Implemented scikit-learn-based anomaly detection to flag unusual spending with clear explanations
- Developed a cross-platform mobile UI (camera-based capture, editable review flow, dashboard with charts) and containerized the local dev environment with Docker

PureGuard - IoT-Based Museum Artifact Protection System | Live Demo *2025*
Group Project | Firebase, Arduino IDE, C++, NodeMCU ESP8266, EasyEDA, JavaScript

- Contributed to a full-stack IoT and embedded systems project, integrating embedded hardware with a real-time cloud application for hardware-software integration, environmental monitoring, and security
- Designed the Firestore database architecture, implemented Firebase Authentication, and built the keypad/solenoid lock and hardware-to-cloud pipeline (UART + WiFi)

SoulBalance - Wellness Blog Platform | Live Demo | GitHub Link *2025*
Individual Project | PHP, MySQL (PDO), JavaScript, HTML5/CSS3, REST API architecture

- Built a full-stack wellness blog platform with auth, blog CRUD via REST API, rich-text editor with image upload, and dual rating system (guest + authenticated)
- Implemented AJAX-powered comments, bcrypt password hashing, and PDO prepared statements against SQL injection

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Active Member- Rotaract Club of University of Moratuwa

- Participated in community service and leadership development initiatives organized by the club, collaborating with members on planning and executing projects that benefited the local community

PUBLICATIONS & WRITING

Tech Blogs on Software Engineering & Backend Development | https://medium.com/@sandunikekulawala

- Authored technical articles on backend systems and applied AI, covering real-world engineering problems, API integration patterns, and production-grade debugging for developers and early-stage professionals. *Examples: LLM Recommendation Engine with Claude , Debugging a Supabase URL Length Bug*

ACHIEVEMENTS & COMPETITION

- Certificate: Python for Beginners - University of Moratuwa (OpenUoM)
- Certificate: Web Design for Beginners- University of Moratuwa (OpenUoM)
- Pursuing AWS Certified Cloud Practitioner
- Participant: MoraXtreme Hackathon 9.0